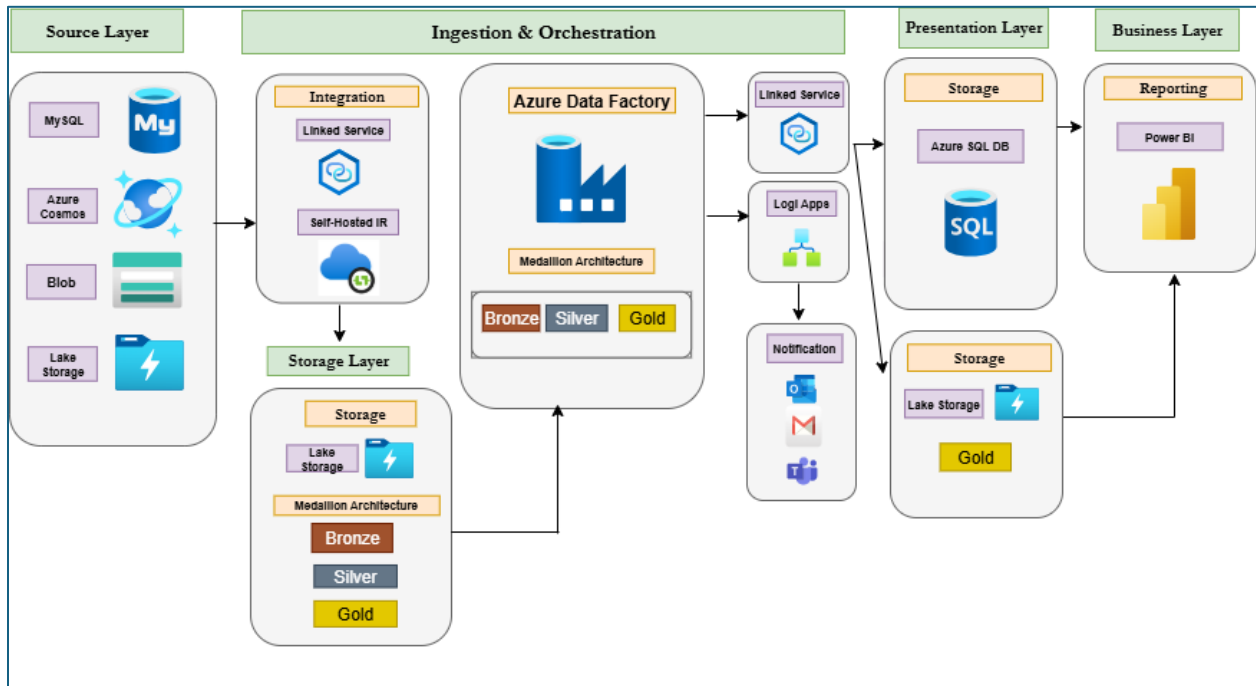


PROJECT OBJECTIVE:

Build a centralized Data Warehousing System in Azure to integrate and transform sales-related data from multiple sources (Flat Files, Excel, CSV, MYSQL, and Cosmos DB). The goal is to provide Sales Insights in a consolidated model for reporting and analytics.

PROJECT ARCHITECTURE:



DATA SOURCES:

- ADLS - Sales (Files in folder by year: csv)
- Blob Container: Categories.csv, SubCategories.csv, SalesRep.xlsx
- Azure Cosmos: Geography.json
- MYSQL: Product_Tbl

DATA MODELING APPROACH:

Fact Table:

- SalesFact

Dimension Tables:

- DimProduct
- DimCategory
- DimSubCategory
- DimGeography
- DimSalesRep

Resources Required:

- Azure Subscription
- Resource Group
- Blob Storage
- Data Lake Storage Gen2
- MySQL
- Azure Cosmos DB
- Azure Data factory
- Logic Apps
- Azure SQLDatabase

Create a Resource Group:

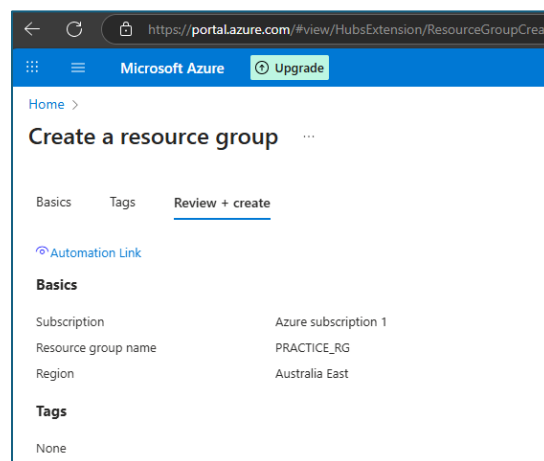
A resource group is like a folder to keep related Azure resources (e.g., storage, databases) organized together.

How to Create a Resource Group:

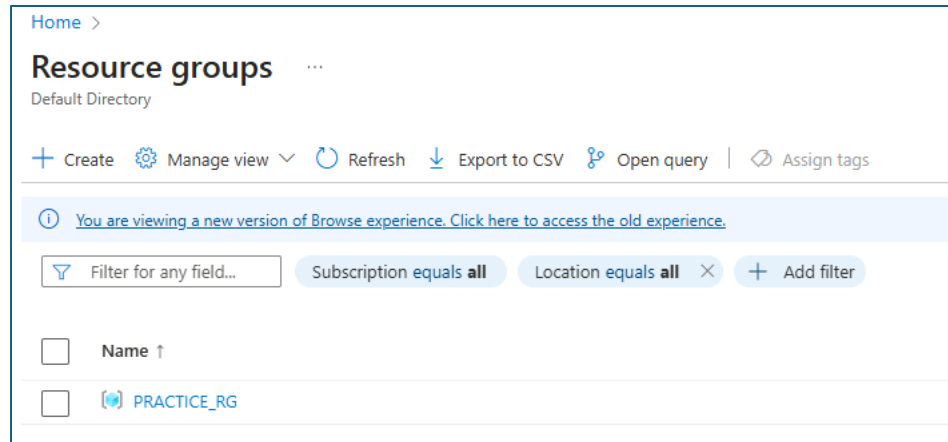
- In the left-hand menu, click “Resource groups”
- Click “+ Create”

Fill in Basic:

- Subscription: Azure Subscription1
- Resource group name: DENIS_RG
- Region: Australia East
- Click Review + Create, then click Create



- ✓ You've now created a resource group!



Source and Target Setup:

1. Azure Blob Storage

Blob storage lives inside something called a storage account — like a big folder in the cloud.

- Container Name: **denisraw**
- Data Format: **CSV / JSON / Parquet**
- Sample File: **Categories.csv, Subcategories.csv, SalesRep.xlsx**
- Storage Account Name: **denissablobs**
- Click "**Create a resource**"
- Search for "**Storage account**" and click Create

1.1 Fill in basic details:

- Subscription: **Azure Subscription1**
- Resource Group: **DENIS_RG**
- Storage account name: **:denissablobs**
- Region: **Australia southeast**
- Primary service: **Azure Blob or Azure Data Lake Storage Gen2**
- Performance: **Standard** (default is fine)
- Redundancy: Choose **Locally-redundant storage (LRS)** for testing or basic use.

1.2 Fill in Advanced details:

- Access Tier: **Hot, Cool, or Cold** based on your usage.

1.3 Fill in Networking Details:

- Network Access: Enable Public access from all networks (for practice).
- Routing Preference : **Microsoft Network Routing.**

1.4 Fill in Data Protection details:

- Soft delete for blobs: **Enable (recommended)**
- Soft delete for containers: **Optional.**
- Soft delete for file shares : **Optional**

1.5 Click **Review + Create**, then **Create**

1.6 Reference:

Basics	
Subscription	Azure subscription 1
Resource group	PRACTICE_RG
Location	Australia East
Storage account name	blob2adf
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard
Replication	Locally-redundant storage (LRS)
Advanced	
Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled
Security	
Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled
Minimum TLS version	Version 1.2
Permitted scope for copy operations (preview)	From any storage account
Networking	
Network connectivity	Public endpoint (all networks)
Default routing tier	Microsoft network routing
Data protection	
Point-in-time restore	Disabled
Blob soft delete	Enabled
Blob retainment period in days	7
Container soft delete	Enabled
Container retainment period in days	7
File share soft delete	Enabled
File share retainment period in days	7

✓ Your storage account is now created!

←

↻

🔒

https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F8954a77b-a278-4750-8e15-4990f3fd498%2FresourceGroup/...

Microsoft Azure

Upgrade

Home >

blob2adf_1752815401246

Overview

...

Deployment

🔍 Search

✕

⏪

🗑️ Delete

⌛ Cancel

🔄 Redeploy

⬇️ Download

🔄 Refresh

👤 Overview

📄 Inputs

📄 Outputs

📄 Template

✔️ Your deployment is complete

📄

Deployment name: blob2adf_1752815401246

Subscription: [Azure subscription 1](#)

Resource group: [PRACTICE_RG](#)

Start time: 7/18/2025, 3:10:29 PM

Correlation ID: 692e4684-5852-4442-941a-d3150f13c6ab [🔗](#)

⌵ Deployment details

⌶ Next steps

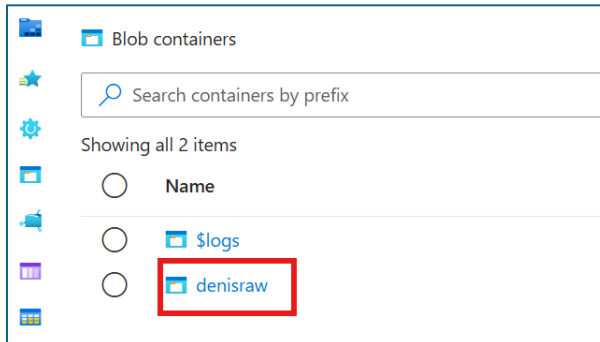
Go to resource

1.7 Create a Blob Container

Containers are like folders inside the storage account to organize files.

- Click "+ Container"
- Give it a name : **denisraw**
- Click Create

1.8 Reference:

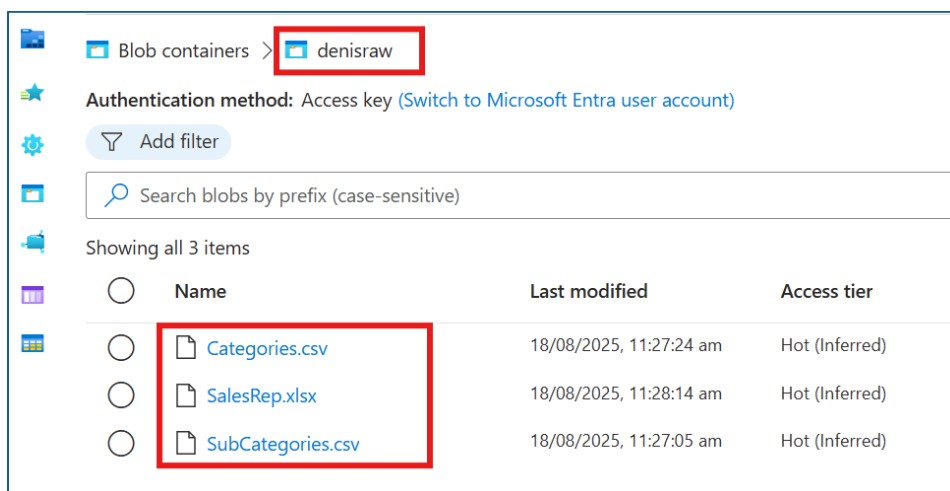


1.9 Upload a Blob (File)

Now, you can upload files (images, videos, logs, etc.):

- Click on the container you just created
- Click "Upload"
- Select a file from your computer
- Click Upload

1.10 Reference:



2. Azure Blob Storage

2.1 Fill in basic details:

1. Subscription: Azure Subscription1
2. Resource Group: DENIS_RG
3. Region: Australia southeast
4. Primary service : Azure Blob or Azure Data Lake Storage Gen2
5. Performance: Standard (default is fine)
6. Redundancy: Choose Locally-redundant storage (LRS) for testing or basic use.

2.2 Fill in Advanced details:

1. Access Tier: Hot, Cool, or Cold based on your usage.
2. Hierarchal Namespace: Enable

Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#) 

Enable hierarchical namespace ⓘ



2.3 Fill in Networking Details:

1. Network Access: Enable Public access from all networks (for practice).
2. Routing Preference : Microsoft Network Routing.

2.4 Fill in Data Protection details:

1. Soft delete for blobs: Enable (recommended)
2. Soft delete for containers: Optional.
3. Soft delete for file shares : Optional

2.5 Click Review + Create, then Create

Blob containers

Search containers by prefix

Showing all 3 items

☐ Name

☐ \$logs

☐ denis-factraw

☐ denis-medallion

Blob containers > denis-factraw > Sales

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive) Only show...

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type
☐	[-]			
☐	Sales 2014.csv	18/08/2025, 11:34:47 am	Hot (Inferred)	Block blob
☐	sales 2015.csv	18/08/2025, 11:34:47 am	Hot (Inferred)	Block blob
☐	sales 2016.csv	18/08/2025, 11:34:50 am	Hot (Inferred)	Block blob
☐	sales 2017.csv	18/08/2025, 11:34:48 am	Hot (Inferred)	Block blob

Medallion Container:

Blob containers > denis-medallion

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Showing all 3 items

☐	Name	Last modified	Access tier	Blob
☐	Bronze	18/08/2025, 11:33:13 am		
☐	Gold	18/08/2025, 11:33:27 am		
☐	Silver	18/08/2025, 11:33:21 am		

3.)CosmosDB:

- Start Cosmos DB Creation
- On the left menu → Click Create a resource.
- Search for Azure Cosmos DB → Select it.
- Click Create.

Basics Tab

- Subscription: **Azure Subscription1**
- Resource Group: **DENIS_RG**
- Region: **Australia southeast**
- Account Name: **Denis**
- Capacity Mode:
 - **Provisioned throughput: Fixed request units (RU/s).**
 - **Serverless: Pay per request (good for dev/test).**

Global Distribution (Optional)

- Cosmos DB supports multi-region replication.
- You can enable Geo-redundancy if you need high availability.

Networking & Security

- Networking tab: **Public endpoint (default) or private endpoint.**
- Backup policy: **Periodic or Continuous backup.**
- Encryption: **Default Microsoft-managed key is fine, or choose Customer-managed key.**

Review + Create

- Click **Review + Create**.
- Validate configuration → Click **Create**.
- Deployment takes a couple of minutes.

Create Database & Container

After deployment:

1. Go to your **Cosmos DB account** in the portal.
2. Under **Data Explorer**, click **New Database**.
 - a. Example: **deniscosmos**.

Load / Query Data

- Upload JSON documents manually via **Data Explorer** or programmatically using SDKs (Python, .NET, Node.js, etc.).
- Example : Geography.json

4.)MYSQL:

CREATING THE IR:

Integration runtime setup

Network environment:

Choose the network environment of the data source / destination or external compute to which the integration runtime will connect to for data flows, data movement or dispatch activities:

**Azure**

Use this for running data flows, data movement, external and pipeline activities in a fully managed, serverless compute in Azure.

**Self-Hosted**

Use this for running data movement, external and pipeline activities in an on-premises / private network by installing the integration runtime.

Note: Data flows are only supported on Azure integration runtime. You can use self-hosted integration runtime to stage the data on cloud storage and then use data flows to transform it.

[View less](#) ^

External Resources:

You can use an existing self-hosted integration runtime that exists in another resource. This way

Continue

Cancel

Integration runtime setup

Private network support is realized by installing integration runtime to machines in the same on-premises network/VNET as the resource the integration runtime is connecting to. Follow below steps to register and install integration runtime on your self-hosted machines.

Name * 

 Validating the integration runtime name...

Description

Type

Self-Hosted

Create

Back

Cancel

Option 1: Express setup

[Click here to launch the express setup for this computer](#)

Option 2: Manual setup

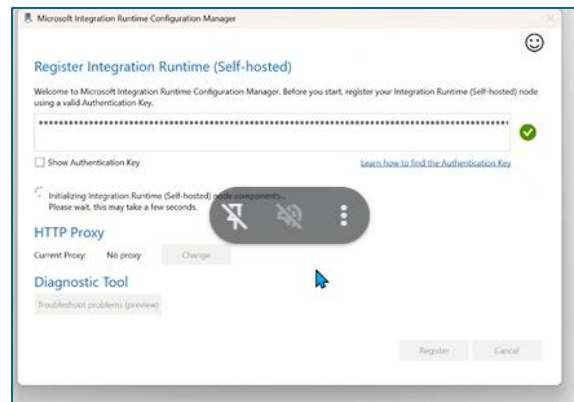
Step 1: **Download and install integration runtime**

Step 2: Use this key to register your integration runtime

Name	Authentication key		
Key1	IR@c260a764-b5b1-4019-966d-31273206756b@onprem2sqladf@Se		
Key2	IR@c260a764-b5b1-4019-966d-31273206756b@onprem2sqladf@Se		

Close

INSTALL MICROSOFT INTEGRATION RUNTIME:



New linked service

MySQL [Learn more](#)

Name *
MySQL1

Description

Connect via integration runtime *
selfhostintegrationRuntime1

Integration runtime selfhostintegrationRuntime1 is currently unavailable. Please try creating linked service once it is online.

Server name *

Port
3306

Database name *

User name *

Password *

SSL mode
Preferred

☐ Use system trust store ☒ Not use system trust store

Additional connection properties
[+ New](#)

Annotations
[+ New](#)

[> Parameters](#)

[> Advanced](#)

New linked service

 MySQL [Learn more](#) 

Name *

MySql1

Description

Connect via integration runtime * 

 selfhostintegrationRuntime1  

Server name *

Port

3306

Database name *

User name *

Password

Azure Key Vault

Password *

SSL mode

Preferred 

☐ Use system trust store ☒ Not use system trust store

Additional connection properties

 New

Annotations

 New

> Parameters

> Advanced 

Query 1 product ×

Limit to 1000 rows

```
1 • SELECT * FROM denis.product;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	ProductID	Sub Category Key	Color	ProductName	RetailPrice	StandardCost
▶	1	3	Red	Alder	23.95	7.55
	2	2	Blue	Linder	23.95	7.55
	3	2	Green	Magnum	23.95	7.55
	4	1	Red	Quad	43.95	13.75

product 1 ×

5. SQLDB: Azure SQL Database

- Database Name: **denisadmin**
- Table Name: **Denis_G**
- Click "**Create a resource**" (top left)
- Search for "**SQL Database**", then click Create

5.1 Fill in the basics:

- Subscription: **Azure Subscription1**
- Resource Group: **DENIS_RG**
- Region: **Australia southeast**
- Database name: **denisdb**
- Server: Click "**Create new**"

Home > Create a resource >

Create SQL Database

Microsoft

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1 ▼

Resource group * ⓘ MYPRACTICE_RG ▼

[Create new](#)

5.2: Create a SQL Server

- This is where your database will live — think of it like the "host" for your database.
- Click Create new (under SQL Server)

5.2.1 Fill in:

- **Server name:** e.g., **denissqlserver** (must be globally unique)
- **Location:** **Australia southeast**
- **Admin login:** Create a username (e.g., sqladmin)
- **Password:** Set a strong password
- Click OK to return to the SQL DB creation form.

Server name * practicedbserverr ✓
 .database.windows.net

Location * (Asia Pacific) Australia East ▼

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login * admin_sql ✓

Password * ✓

Confirm password * ✓

✓ Password and confirm password must match.

5.3: Configure the Database Settings

- You can leave defaults, or:
 - Choose Standard or Basic pricing tier (for low-cost testing)
 - Click Configure database if you want to adjust size or performance

Configure

[Feedback](#)

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

☒ SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier: Basic (For less demanding workloads) [Compare service tiers](#)

DTUs: [Compare DTU options](#)

5 (Basic)

Data max size (GB)

[Apply](#)



Cost summary

Basic (Basic)	
Cost per DTU (in USD)	1.08
DTUs selected	x 5
ESTIMATED COST / MONTH	5.39 USD

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

- Backup storage redundancy ⓘ
- ☒ Locally-redundant backup storage
 - ☐ Zone-redundant backup storage
 - ☐ Geo-redundant backup storage
 - ☐ Geo-Zone-redundant backup storage

5.4: Enable Public Access (Firewall Rule)

To access your database from your laptop:

Connectivity Method: **Public Endpoint**

- In the Networking tab, under Firewall rules, select:
- Allow Azure services and resources to access this server: **yes**
 - **Add your current client IP address**

- This lets you connect to the database from your machine

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method * ⓘ

- ☐ No access
- ☒ Public endpoint
- ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

5.5: Review and Create

- Click Review + create
- Wait for validation
- Click Create

Microsoft.SqlDatabase.newDatabaseNewServer_8217e5e82f2e4d9394264 | Overview

Deployment

×
<<

Delete
 Cancel
 Redeploy
 Download
 Refresh

Overview

Inputs

Outputs

Template

Deployment name : Microsoft.SqlDatabase.new...
 Start time : 7/17/2025, 5:10:31 PM

Subscription : Azure subscription 1
 Correlation ID : 30e77e59-69d7-439e-983f-1...

Resource group : MYPRACTICE_RG

Deployment details

Resource	Type	Status	Operations
practicesqlservr/...	Microsoft.Sql/servers/firewallrul	OK	Operations
practicesqlservr/...	Microsoft.Sql/servers/firewallrul	OK	Operations
practicesqlservr...	Microsoft.Sql/servers/databases	Accepted	Operations
practicesqlservr/l	Microsoft.Sql/servers/connection	OK	Operations
practicesqlservr	Microsoft.Sql/servers	Created	Operations

Microsoft.SQLDatabase.newDatabaseNewServer_8d6663fecf36436bbaf33 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_8d6663fecf36436bbaf33

Subscription : [Azure subscription 1](#)

Resource group : [PRACTICE_RG](#)

> Deployment details

< Next steps

Go to resource

practicesqlldb (practicesqlserverr/practicesqlldb) | Query editor (preview)

SQL database

Search

Login New Query Open query Feedback Getting started

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Query editor (preview) is a tool to run SQL queries against Azure SQL Database in the Azure portal. It is designed for lightweight querying and object exploration in your database. For more information and troubleshooting, [Learn more](#)

SQL

Welcome to SQL Database Query Editor

SQL server authentication

Microsoft Entra authentication

Login

admin_sql

Password

Logging in as admin_sql...

OR

Continue as mahalakshmi101018@outlo...

OK

6.) Create Azure Data Factory (ADF)

6.1 Start Creating Data Factory

1. Click **Create a resource** (top-left).
2. Search for **Data Factory**.
3. Select **Data Factory** → Click **Create**.

Basics Tab

- Subscription: **Azure Subscription1**
- Resource Group: **DENIS_RG**
- Region: **Australia southeast**
- Name: **denisadftest**
- Version: Select **V2 (Latest)**.

Review + Create

- Validate settings.
- Click **Create**.

6.2 Steps Inside ADF Studio

Create Linked Services (Connections)

- Linked Services = connection strings to external data sources.
- Examples for our project:
 - **ADLS Gen2** (for raw/staging/curated zones).
 - **Azure SQL Database** (for target warehouse).
 - **File System / Blob / On-prem DB** (if required).

Create Datasets

- Dataset = Representation of a data structure (like a table or file).

- Examples:
 - Sales_2023.csv (CSV file in ADLS).
 - DimProduct (SQL table).

Create Pipelines

- Pipelines = Set of activities (ETL flow).
- Typical pipeline tasks:
 - **Copy Activity** → Move raw files from source → ADLS Raw Zone.
 - **Data Flow / Mapping Data Flow** → Transform data (split Location, clean IDs, create GeoKey).
 - **Copy Activity** → Load curated tables into **Azure SQL/Synapse**.
 - **Validation** → Check row counts / success.

Raw to Bronze Layer

Pipeline1: raw2bronze_PL1

Dimension (category,salesrep,subcategory from blob to adls bronze folder.

Source: Blob Storage

Target: ADLS Bronze

New linked service

Azure Blob Storage [Learn more](#)

Name *
blob2adf

Description

Connect via integration runtime *
 AutoResolveIntegrationRuntime

Authentication type
Account key

Connection string **Azure Key Vault**

Account selection method
☒ From Azure subscription ☐ Enter manually

Azure subscription

Connection successful
 Test connection

Create **Cancel**

Source:

Data Factory Validate all Publish all 2

AzureSqlTable1 raw2bronze_PL1

Validate Validate copy runtime Debug Add trigger

General **Source** Sink Mapping Settings User properties

Source dataset * rawfiles

Open New Preview data [Learn more](#)

File path type ☐ File path in dataset ☐ Prefix ☒ Wildcard file path ☐ List of files

Wildcard paths denisraw / Wildcard folder path

*

[Add dynamic content \[Alt+Shift+D\]](#)

Start time (UTC) End time (UTC)

Sink:

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

data

< All Azure Database File Generic protocol >

 Azure Data Explorer (Kusto)	 Azure Data Lake Storage Gen2	 Azure Database for MySQL
		

Continue Cancel

Set properties

Name
bronzefiles

Linked service *
bronzeputls

File path
denis-medallion / Bronze / File name

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

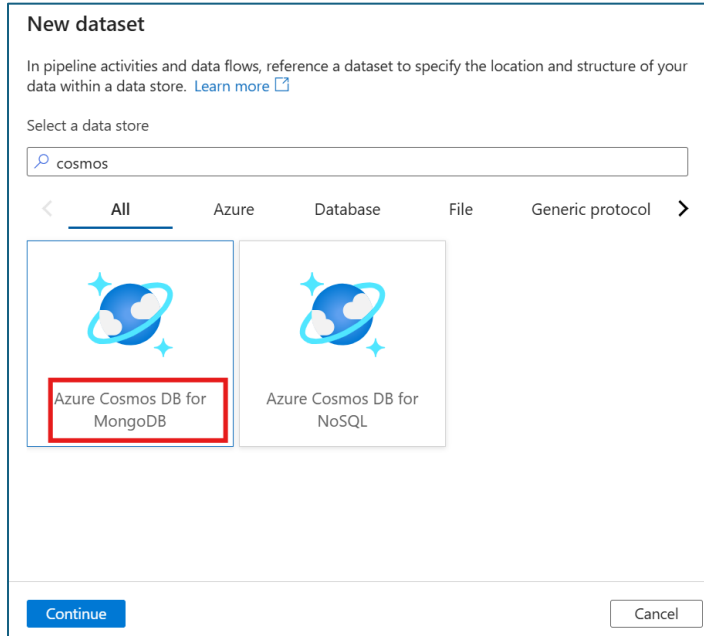
> Advanced

OK Back Cancel

Pipeline2: *onprem2bronze PL2*

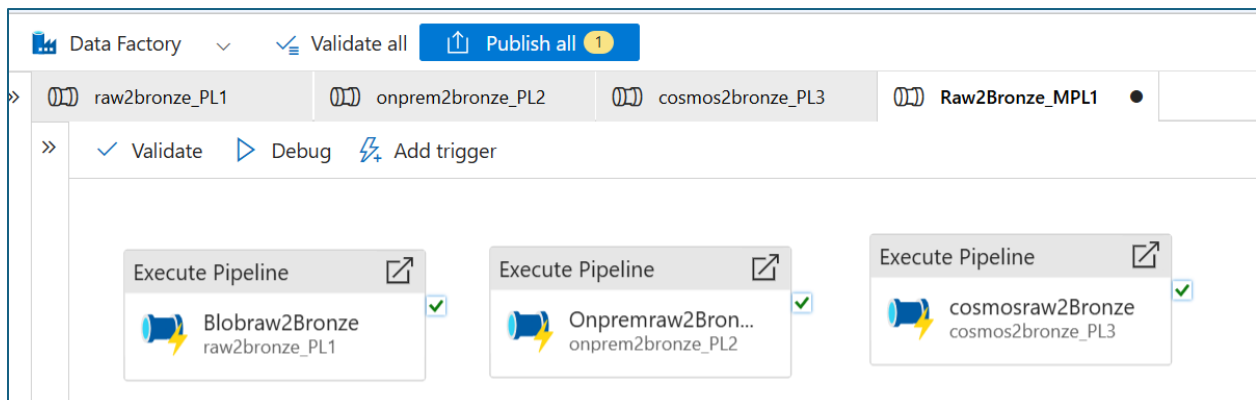
Source: OnpremInputData

Target: OnpremBronzeFiles



Main PL:

Raw2Bronze_MPL1



New linked service

Azure Cosmos DB for MongoDB [Learn more](#)

✓ AutoResolveIntegrationRuntime

Connection string Azure Key Vault

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

Azure subscription 1 (07cad030-cb00-4648-a341-2f026c59f324)

Azure Cosmos DB account name * ⓘ

deniscosmosdbtest

Database name *

denisdb

Annotations

+ New

> Advanced ⓘ

Create **Cancel** Test connection

Files moved from raw to bronze:

Here all the files are having different names and format, we need to fix it.

Blob containers > denis-medallion > Bronze

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive) Only show active objects

Showing all 9 items

	Name	Last modified	Access tier	Blob type
	Categories.csv	19/08/2025, 9:56:22 am	Hot (Inferred)	Block blob
	Sales 2014.csv	19/08/2025, 9:56:23 am	Hot (Inferred)	Block blob
	SalesRep.xlsx	19/08/2025, 9:56:22 am	Hot (Inferred)	Block blob
	SubCategories.csv	19/08/2025, 9:56:22 am	Hot (Inferred)	Block blob
	`product`.txt	19/08/2025, 9:57:02 am	Hot (Inferred)	Block blob
	data_55d7c9ca-60d8-4f9b-9290-f0495c54e6d9_ee44a8...	19/08/2025, 9:56:22 am	Hot (Inferred)	Block blob

Fix 1:

Sales file should go to the Sales folder in the Bronze Layer.

The screenshot shows the Azure Data Studio interface. On the left, a list of datasets includes 'cosmos2bronze_PL3', 'onprem2bronze_PL2', 'raw2bronze_PL1', 'Raw2Bronze_MPL1', 'Change Data Capture (preview)', and 'Datasets' (8). The 'Sales_B' dataset is selected and highlighted with a red box. The main panel shows the 'Connection' tab for this dataset. It includes a 'Linked service' dropdown set to 'bronzeoutputls', a 'File path' field containing 'denis-medallion / Bronze/Sales / File name' (highlighted with a red box), and options for 'Compression type' (No compression), 'Column delimiter' (Comma (,)), and 'Row delimiter' (Default (\r\n, or \n)).

Fix2:

Product & Geo file should have as follow.

- Product.csv
- Geo.json

The screenshot shows the Azure Data Studio interface. On the left, a list of datasets includes 'onprem2bronze_PL2' (highlighted with a red box), 'raw2bronze_PL1', and 'Raw2Bronze_MPL1'. The main panel shows the 'Connection' tab for this dataset. It includes a 'Linked service' dropdown set to 'onpremoutputdata', a 'File path' field containing 'denis-medallion / Bronze / Product.csv' (highlighted with a red box), and options for 'Compression type' (No compression), 'Column delimiter' (Comma (,)), and 'Row delimiter' (Default (\r\n, or \n)).

The screenshot shows the Azure Data Studio interface. On the left, a list of datasets includes 'cosmos2bronze_PL3' (highlighted with a red box), 'onprem2bronze_PL2', 'raw2bronze_PL1', and 'Raw2Bronze_MPL1'. The main panel shows the 'Connection' tab for this dataset. It includes a 'Linked service' dropdown set to 'bronzeoutputls', a 'File path' field containing 'denis-medallion / Bronze / Geo.json' (highlighted with a red box), and options for 'Compression type' (No compression), 'Column delimiter' (Comma (,)), and 'Row delimiter' (Default (\r\n, or \n)).

After Fix:

Blob containers > denis-medallion > Bronze > Sales

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 4 items

	Name	Last modified	Access tier	Blob type
	[.]			
	Sales 2014.csv	19/08/2025, 10:08:45 am	Hot (Inferred)	Block blob
	sales 2015.csv	19/08/2025, 10:08:45 am	Hot (Inferred)	Block blob
	sales 2016.csv	19/08/2025, 10:08:45 am	Hot (Inferred)	Block blob
	sales 2017.csv	19/08/2025, 10:08:45 am	Hot (Inferred)	Block blob

Blob containers > denis-medallion > Bronze

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 6 items

	Name	Last modified	Access tier	Blob type
	[.]			
	Sales	19/08/2025, 10:04:50 am		
	Categories.csv	19/08/2025, 10:08:46 am	Hot (Inferred)	Block blob
	Geo.json	19/08/2025, 10:08:44 am	Hot (Inferred)	Block blob
	Product.csv	19/08/2025, 10:08:47 am	Hot (Inferred)	Block blob
	SalesRep.xlsx	19/08/2025, 10:08:46 am	Hot (Inferred)	Block blob

Bronze to Silver Layer

- From the bronze layer we have to read the file one by one.
- Need to clean the file then push it to silver Layer.

Transformation :

1.) Category.csv

Create a dataflow:

1.)category_DF

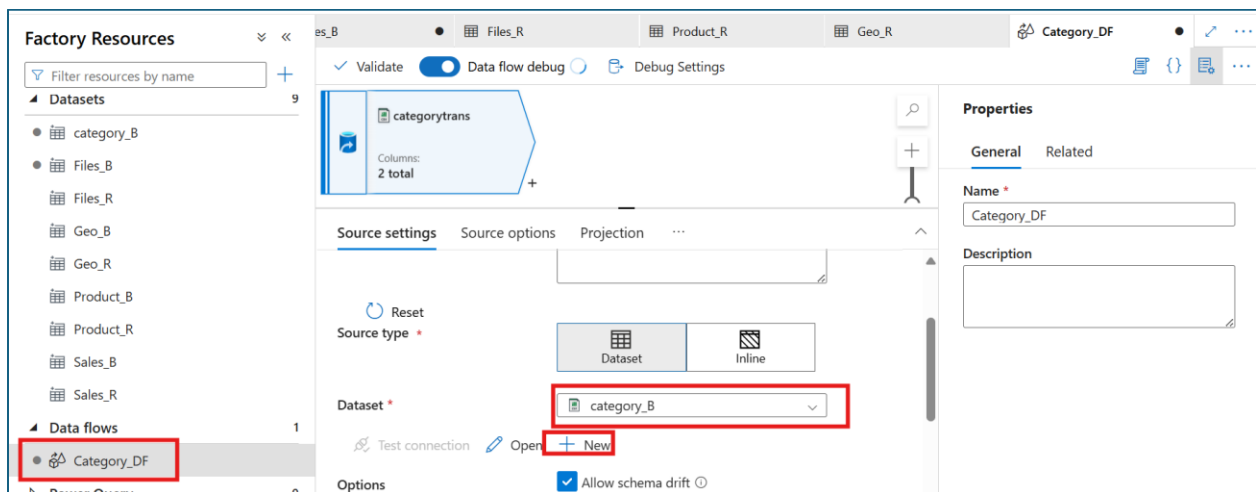
Source: ADLS :

Create Dataset:

1.)category_B

Sink:

1.) from bronze-->category.csv



2.)Subcategory.csv

Create a dataflow:

1.)subcategory_DF

Source: ADLS :

Create Dataset:

1.)subcategory_B

Sink:

1.) from bronze-->subcategory.csv

Set properties

Name

Linked service *

File path
 / /

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

> Advanced

Data cleaning:

1. Need to remove the ID- from the categorykey,

a. Create a Calculated Column

Dataflow expression builder Expression reference documentation

RemoveID

Derived Columns
 Create new

abc Subcategorykeynew

1

Column name *

Expression

2

Expression elements
All
Functions

Expression values

 Create new
abc SubCategoryKey

Data preview Refresh

Save and finish

Cancel

Clear contents

3

But Right function will not work all the time, so the workaround should be

Method1: using substring()

`substring(SubCategoryKey, 5, 9999)`

Method2: Extract only numbers

Source settings			Source options			Data preview		
of rows			INSERT	4	UPDATE	0	DELETE	0
			UPSERT	0	LOOKUP	0	ERROR	0
resh			Typecast	▼	Modify	▼	Map drifted	
SubCategoryKey			abc	↑↓	CategoryKey	abc	↑↓	SubCategory Name
1			ID - 1			Extra		
2			ID - 2			Regular		
3			ID - 1			Micro		
4			ID - 2			Super		

Run the Pipeline

Data Factory | Validate all | Publish all | Preview experience | Off

Product_R | Geo_R | category_B | category_DF | subcategory_DF | **Bronze2Silver**

Validate | Debug | **Add trigger** | Data flow debug | On

Data flow | Category | Subcategory

General | Settings | Parameters | User properties

Data flow * | subcategory_DF | Open | New

Properties

General | Related

Name * | **Bronze2Silver**

Description

Annotations

Output:

Rerun | Cancel | Refresh | Update pipeline | List | Gantt

Data flow | Category

Activity runs

Pipeline run ID 5a32f5a7-ebb5-4e46-af62-6f44a7d73ef1

All status | Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Subcategory	✓ Succeeded	Data flow	8/19/2025, 12:51:21 PM	3m 8s	AutoResolveIntegrationRuntime (Australia So
Category	✓ Succeeded	Data flow	8/19/2025, 12:51:21 PM	4m 47s	AutoResolveIntegrationRuntime (Australia So

Output Validation: (we need to handle the filenames in the later part)

Blob containers > denis-medallion > Silver

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 3 items

	Name	Last modified	Access tier
☐	[.]		
☐	_SUCCESS	19/08/2025, 12:56:05 pm	Hot (Inferred)
☐	part-00000-39d29109-ac69-4819-a...	19/08/2025, 12:56:05 pm	Hot (Inferred)
☐	part-00000-9aea1775-9700-4016-b...	19/08/2025, 12:54:24 pm	Hot (Inferred)

3.) Geo.csv

Steps Implemented:

1. Create a dataflow: geo_DF
2. Use a existing Dataset : Geo_B
3. Create a surrogate key: because we have asked to create a unique key column.

✓ Validate Data flow debug Debug Settings

getgeo Import data from Geo_B

surrogateKey1 Adding new key GeoK starting from 1 with step 1

TargetSilver Columns: 4 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name * TargetSilver Learn more

Description Export data to Geo_S Reset

Incoming stream * surrogateKey1

Sink type * Dataset Inline Cache

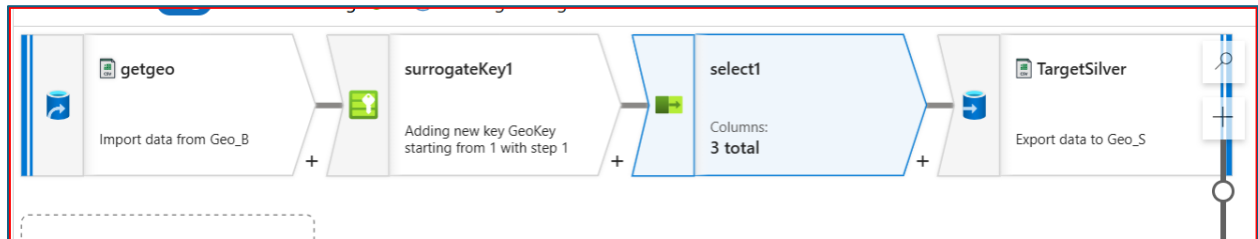
Remove wikipedia column and rearrange the columns as follows:

Input columns *

☐ Auto mapping ⓘ Reset + Add mapping Delete 3 mappings: 1 column(s) from the

☐	surrogateKey1's column	Filter	Name as	Filter
☐	12l GeoKey	→	GeoKey	
☐	abc Country	→	Country	
☐	abc Town	→	Town	

Final Dataflow:



4.) Product.csv

- Create a dataflow: Product_DF
- Use existing Dataset: Product_B
- Do Import Projection because we are not getting any columns.

Remove the duplication:

✓ Validate Data flow debug Debug Settings

Source settings	Source options	Projection	Optimize	Inspect	Data preview	
+	2	2	Blue	Linder	23.95	7.55
+	3	2	Green	Maqnum	23.95	7.55
+	4	1	Red	Quad	43.95	13.75
+	5	1	Blue	Black Monk	43.95	13.75
+	6	4	Green	Quad	43.95	13.75
+	7	1	Red	Binq	26.95	8.25
+	8	3	Blue	VanHelen	26.95	8.25
+	9	1	Green	Maqnum	26.95	8.25
+	10	1	Florescent Pink	Carlota	29.95	9.15
+	11	4	Florescent Blue	Carlota	29.95	9.15
+	7	1	Red	Binq	26.95	8.25

We need to find which one has duplicate.

Aggregate---->Group by

Group by		Aggregates	
Columns		Name as	
12s	ProductID	ProductID	+
abc	ProductName	ProductName	+
12s	Sub Category Key	Sub Category Key	+
abc	Color	Color	+
1.2	RetailPrice	RetailPrice	+
1.2	StandardCost	StandardCost	+

Aggregate--->Aggregates

count(ProductID)

13 rows became 11 rows:

Aggregate settings

Optimize

Inspect

Data preview

Number of rows

INSERT 11

UPDATE 0

DELETE 0

UPSERT 0

LOOKUP 0

ERR

Refresh

Typecast

Modify

Map drifted

Statistics

Remove

Export to CSV

	ProductID 12s	ProductName abc	Sub Category Key 12s	Color abc	RetailPrice 1.2	StandardCost 1.2
+	1	Alder	3	Red	23.95	7.55
+	2	Linder	2	Blue	23.95	7.55
+	3	Magnum	2	Green	23.95	7.55
+	4	Quad	1	Red	43.95	13.75

Remove the repeated columns: (here :countrows)

Select settings Optimize Inspect Data preview ●

Input columns *

☐ Auto mapping ⓘ  Reset  Add mapping  Delete 6 mappings: 1 column(s) from the

☐	aggregate1's column		Name as
☐	12s ProductID	→	ProductID
☐	abc ProductName	→	ProductName
☐	12s Sub Category Key	→	Sub Category Key
☐	abc Color	→	Color
☐	1.2 RetailPrice	→	RetailPrice
☐	1.2 StandardCost	→	StandardCost

5.) SalesRep.xlsx

Source:

Set properties

Name

Linked service *
 

File path
 / /  | 

Worksheet mode
☒ Name ☐ Index

Sheet name * ⓘ
 

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

> Advanced

Remove the ID – 1 to 1

Follow the previous method.

6.) Sales

- All the files are coming in the folder.
- Create a dataflow
- use the existing dataset : Sales_B

Sink : Sales_S

GetSalesdata--->Import Schema

Split the columns: location into Country & State

Country:

```
split(Location, ';')[1]
```

State:

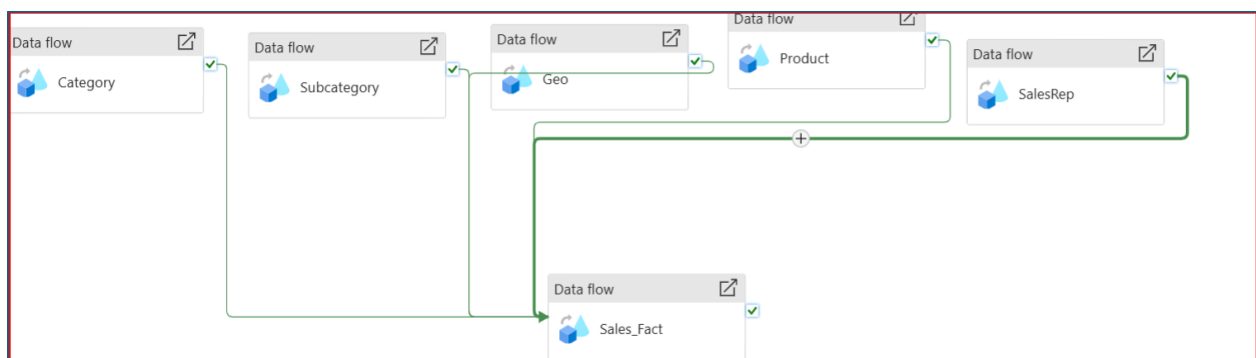
```
split(Location, ';')[2]
```

Date : change the format of the Date column as below

```
toDate(Date, 'dd.MM.yyyy')
```

Run the pipeline:

Dimension first then followed by Fact Table.



All status ▾

[Monitor in Azure Metrics](#) [Export to CSV](#) ▾

Showing 1 - 6 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Sales_Fact	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	2m 46s	AutoResolveIntegrationRuntime (Australia So
Geo	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	3m 16s	AutoResolveIntegrationRuntime (Australia So
Subcategory	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	2m 50s	AutoResolveIntegrationRuntime (Australia So
SalesRep	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	2m 53s	AutoResolveIntegrationRuntime (Australia So
Category	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	2m 57s	AutoResolveIntegrationRuntime (Australia So
Product	✓ Succeeded	Data flow	8/19/2025, 5:15:06 PM	2m 49s	AutoResolveIntegrationRuntime (Australia So

Silver Folder:

	Name	Last modified	Access tier	Blob type	Size	Lease state
○	part-00000-722fd28a-f9d9-4155-8934-7a3d...	19/08/2025, 5:17:48 pm	Hot (Inferred)	Block blob	916.68 KiB	Available
○	part-00000-87fbc45-3228-41e9-adf6-f3f33...	19/08/2025, 5:17:51 pm	Hot (Inferred)	Block blob	395 B	Available
○	part-00000-ba5c32c7-ea2b-4f0b-a8a6-c52a...	19/08/2025, 5:18:18 pm	Hot (Inferred)	Block blob	167 B	Available
○	part-00001-722fd28a-f9d9-4155-8934-7a3d...	19/08/2025, 5:17:48 pm	Hot (Inferred)	Block blob	642.52 KiB	Available
○	part-00002-722fd28a-f9d9-4155-8934-7a3d...	19/08/2025, 5:17:48 pm	Hot (Inferred)	Block blob	627.3 KiB	Available
○	part-00003-722fd28a-f9d9-4155-8934-7a3d...	19/08/2025, 5:17:48 pm	Hot (Inferred)	Block blob	619.59 KiB	Available

Why am i getting these many files?

Also need to change the name of the file too..

WA:

while writing it back -->settings--->Filename option: from Default to Pattern as Category_S

Ex:

getgeo

surrogateKey1

select1

TargetSilver

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Clear the folder

File name option *

Pattern *

Quote All

Pattern

Geo_S

Add dynamic content [Alt+Shift+D]

Sink Settings Errors Mapping **Optimize** Inspect Data preview ●

i This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended partitioning.

Partition option * ☐ Use current partitioning ☒ **Single partition** ☐ Set partitioning

After Fix the Silver Layer:

Blob containers > denis-medallion > **Silver**

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

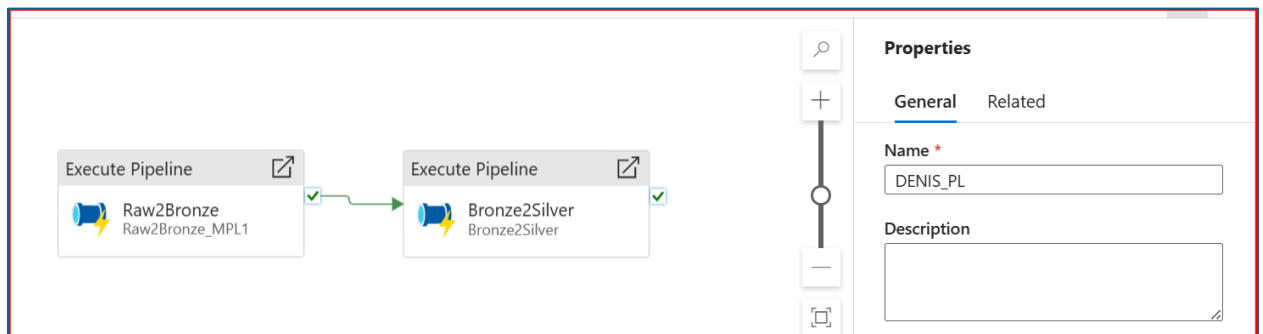
Search blobs by prefix (case-sensitive)

Showing all 6 items

	Name	Last modified	Access tier
☐	Category_S	19/08/2025, 5:35:25 pm	Hot (Inferred)
☐	Geo_S	19/08/2025, 5:35:23 pm	Hot (Inferred)
☐	Product_S	19/08/2025, 5:35:21 pm	Hot (Inferred)
☐	SalesRep_S	19/08/2025, 5:35:29 pm	Hot (Inferred)
☐	Sales_S	19/08/2025, 5:35:56 pm	Hot (Inferred)
☐	Subcategory_S	19/08/2025, 5:35:21 pm	Hot (Inferred)

Final Pipeline: DENIS_PL

- Add the Execute Pipeline for Raw2Bronze, Bronze2Silver.



Bronze to Silver will run after the success run of Raw2Bronze.

[All pipeline runs](#) > **DENIS_PL - Activity runs**

Rerun Cancel Refresh Update pipeline List Gantt

Execute Pipeline

Raw2Bronze
Raw2Bronze_MPL1

→

Bronze2Silver
Bronze2Silver

Execute Pipeline

Activity runs

Pipeline run ID 359db113-85e5-4a87-931d-6fa94b37c405

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Raw2Bronze	In progress	Execute Pipelin	8/20/2025, 10:07:17 AM	3s	

Execute Pipeline

Raw2Bronze
Raw2Bronze_MPL1

→

Bronze2Silver
Bronze2Silver

Execute Pipeline

Activity runs

Pipeline run ID 359db113-85e5-4a87-931d-6fa94b37c405

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Bronze2Silver	In progress	Execute Pipelin	8/20/2025, 10:07:43 AM	43s	
Raw2Bronze	Succeeded	Execute Pipelin	8/20/2025, 10:07:17 AM	26s	

Output:

<

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

...

🗂 Blob containers >

🗂 denis-medallion >

📁 Bronze

Authentication method: Access key [\(Switch to Microsoft Entra user account\)](#)

🔍 Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 6 items

☐	Name	Last modified	Access tier	Blot
☐	📁 Sales	19/08/2025, 10:04:50 am		
☐	📄 Categories.csv	20/08/2025, 10:07:36 am	Hot (Inferred)	Bloc
☐	📄 Geo.json	20/08/2025, 10:07:35 am	Hot (Inferred)	Bloc
☐	📄 Product.csv	20/08/2025, 10:07:35 am	Hot (Inferred)	Bloc
☐	📄 SalesRep.xlsx	20/08/2025, 10:07:36 am	Hot (Inferred)	Bloc
☐	📄 SubCategories.csv	20/08/2025, 10:07:36 am	Hot (Inferred)	Bloc

Bronze2Silver:

<

+ Add Directory

↑ Upload

↻ Refresh

|

🗑 Delete

📄 Copy

📄 Paste

⋮

🗂 Blob containers

>

🗂 denis-medallion

>

📁 Silver

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

🔍

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 6 items

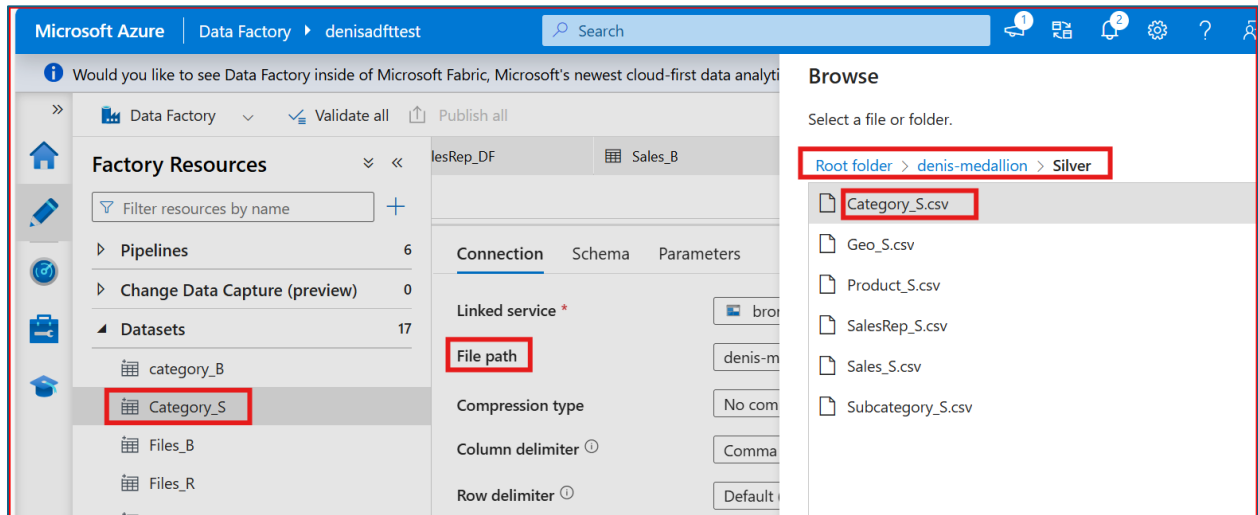
	Name	Last modified	Access tier	Blob type
	📄 Category_S...	20/08/2025, 10:10:31 am	Hot (Inferred)	Block blob
	📄 Geo_S.csv	20/08/2025, 10:10:34 am	Hot (Inferred)	Block blob
	📄 Product_S.csv	20/08/2025, 10:10:31 am	Hot (Inferred)	Block blob
	📄 SalesRep_S...	20/08/2025, 10:10:40 am	Hot (Inferred)	Block blob
	📄 Sales_S.csv	20/08/2025, 10:11:04 am	Hot (Inferred)	Block blob
	📄 Subcategor...	20/08/2025, 10:10:32 am	Hot (Inferred)	Block blob

Silver to Gold Layer

While reading in Gold Layer from Silver, it may not know the file name, as we didn't give the filename during writing into the silver layer. For that, follow the below steps.

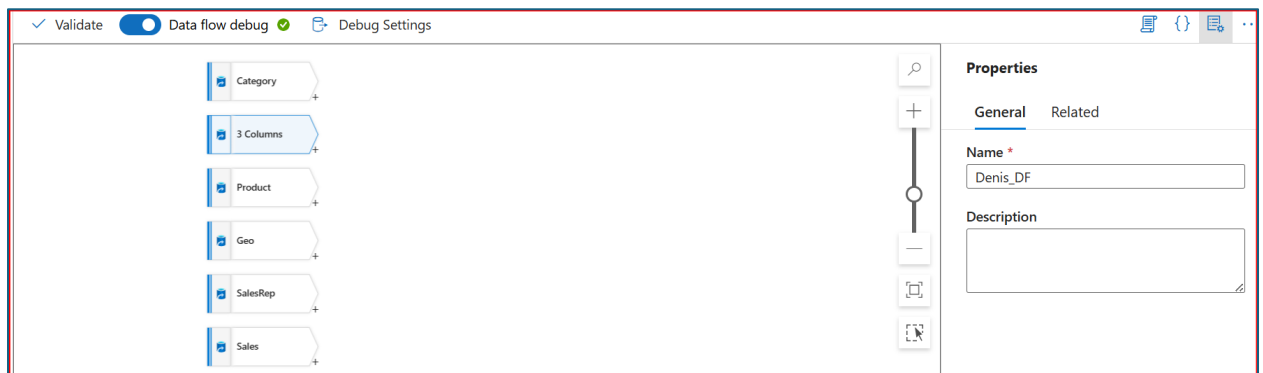
Repeat for all the Datasets.

Ex: category.csv



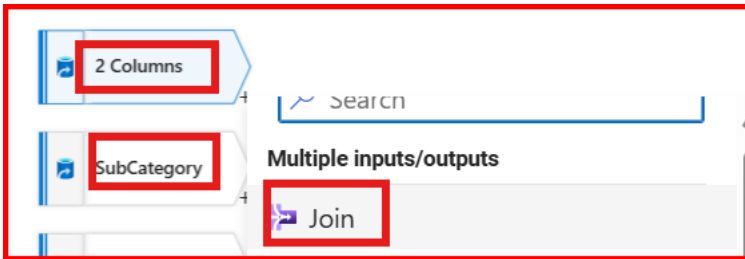
Create a Dataflow :Denis_DF to read the data from silver file and merge into a single file (Denis_G.csv)

Do the Import Projection for all the Dataset to see the schema.



Now we need to create a single File to give the Business Intelligence team.

Bring Category and Subcategory into single file by joining.



Category_S Product_S Sales_S SalesRep_S Subcategory_S

✓ Validate Data flow debug ✓ Debug Settings

Join settings Optimize Inspect Data preview ●

← Previous Next →

Output stream name * CategorySubcategory [Learn more](#)

Description Inner join on 'Category' and 'SubCategory' [Reset](#)

Left stream * Category

Right stream * SubCategory

Join type *

Full outer Inner Left outer Right outer Custom (cross)

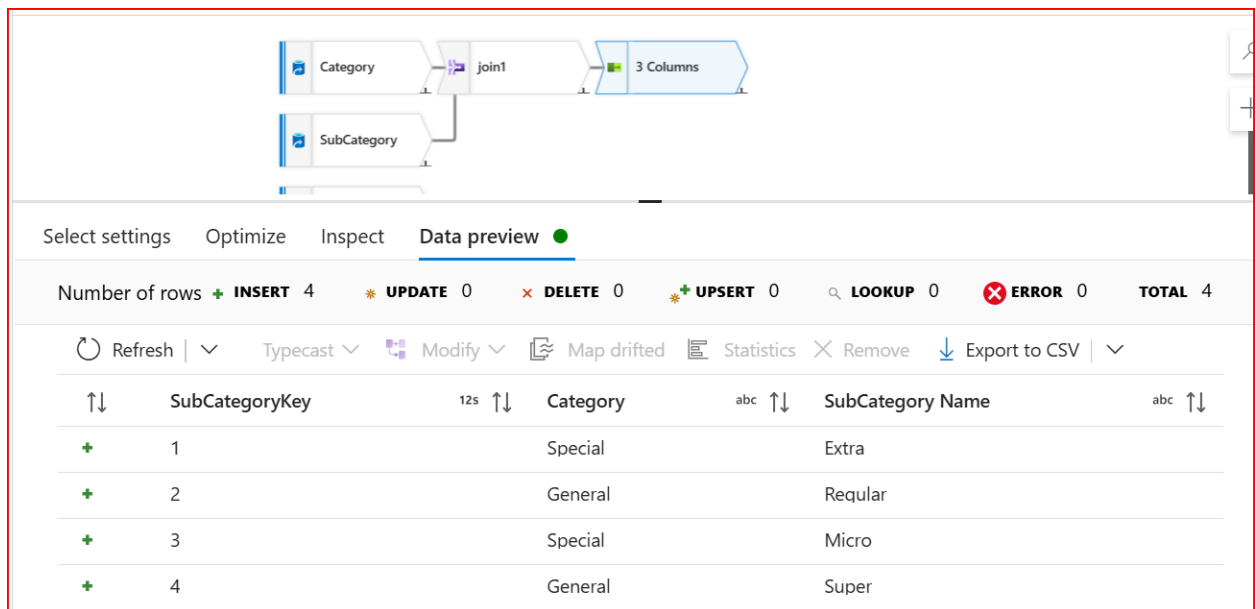
Use fuzzy matching ⓘ ☐

Join conditions *

Left: Category's column Right: SubCategory's column

12s CategoryKey == 12s Categorykey +

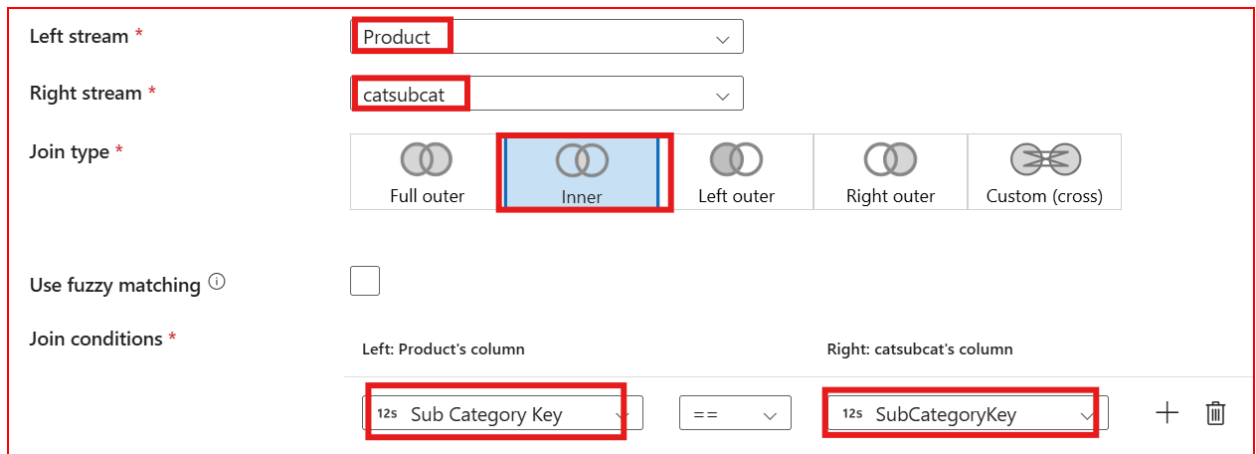
Output:



The screenshot shows a data preview of a join operation. The top part displays a visual workflow with 'Category' and 'SubCategory' streams joined at 'join1', resulting in '3 Columns'. Below this, the 'Data preview' tab is active, showing a summary of row counts: INSERT 4, UPDATE 0, DELETE 0, UPSERT 0, LOOKUP 0, ERROR 0, and TOTAL 4. A toolbar includes options like Refresh, Typecast, Modify, Map drifted, Statistics, Remove, and Export to CSV. The main table displays the joined data with columns: SubCategoryKey, Category, and SubCategory Name.

↑↓	SubCategoryKey	12s ↑↓	Category	abc ↑↓	SubCategory Name	abc ↑↓
+	1		Special		Extra	
+	2		General		Regular	
+	3		Special		Micro	
+	4		General		Super	

Now Join **Product** with **Catsubcat** dataset: procatsubcat



The screenshot shows the configuration for joining 'Product' and 'catsubcat' streams. The 'Left stream' is 'Product' and the 'Right stream' is 'catsubcat'. The 'Join type' is set to 'Inner'. The 'Use fuzzy matching' checkbox is unchecked. The 'Join conditions' section shows 'Left: Product's column' as '12s Sub Category Key' and 'Right: catsubcat's column' as '12s SubCategoryKey', with an equals sign operator between them.

Left stream * **Product**

Right stream * **catsubcat**

Join type * **Inner**

Use fuzzy matching ☐

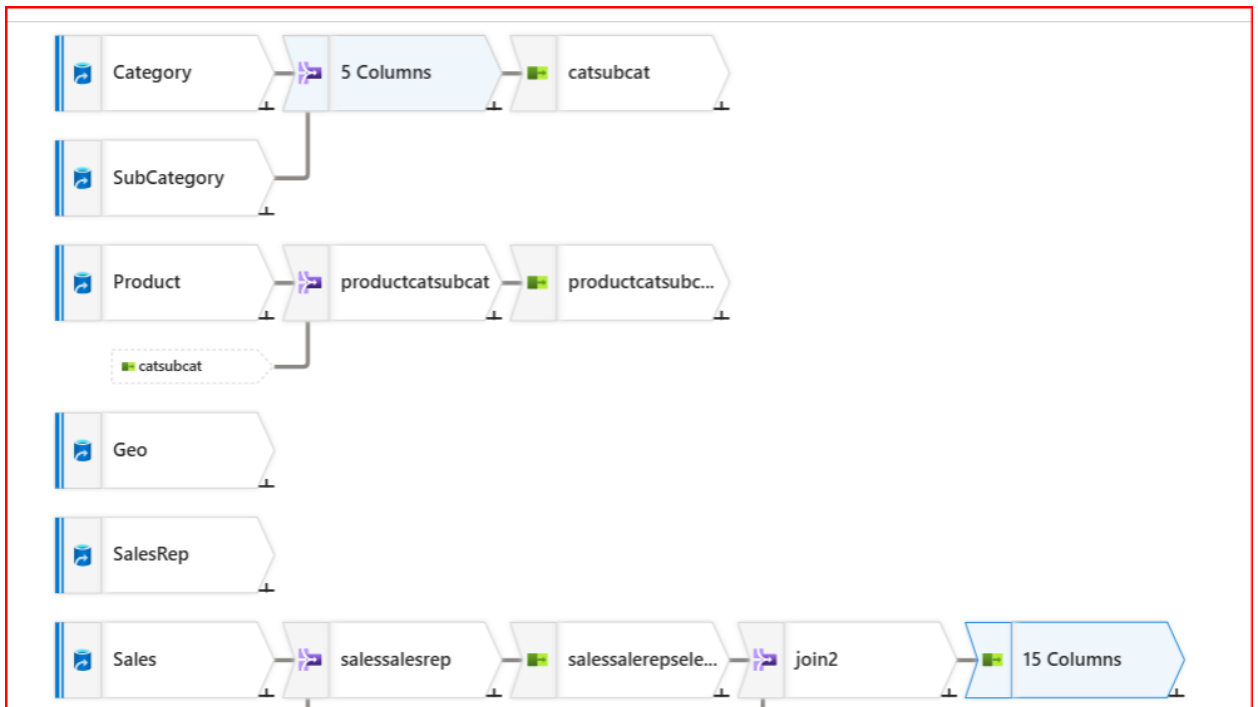
Join conditions *

Left: Product's column **12s Sub Category Key** == **12s SubCategoryKey** Right: catsubcat's column

Join sales and salesrep: salessalesrepselect

Join procatsubcat and salessalesrepselect

Final Dataflow:



Calculate :

Total_Revenue

Column name *
TotalRevenue
Expression
RetailPrice*Units

Column name *
TotalCost
Expression
costStandardCost*Units

Column name *

GrossProfit

Expression

TotalRevenue - TotalCost

Create a new pipeline for Silver2Gold:

Data Factory

Validate all

Publish all

Product_S

Sales_S

SalesRep_S

Subcategory_S

Denis_DF

Validate

Debug

Add trigger

Data flow debug

Data flow

Silver2Gold

Update the Existing Main Pipeline Denis_PL to call the Silver2Gold .

Validate

Debug

Add trigger

Data flow debug

Execute Pipeline

Raw2Bronze

Raw2Bronze_MPL1

Execute Pipeline

Bronze2Silver

Bronze2Silver

Execute Pipeline

Silver2Gold

Silver2Gold

Output:

All pipeline runs

Silver2Gold - Activity runs

Rerun

Cancel

Refresh

Update pipeline

List

Gantt

Data flow

Silver2Gold

<

+ Add Directory

↶ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

⋮

🗂 Blob containers

>

🗂 denis-medallion

>

📁 Gold

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

🔍

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type
☐	📁 [..]			
☐	📄 Denis_G.csv	20/08/2025, 12:44:52 pm	Hot (Inferred)	Block blob

We can also write the Gold data into a Azure SQLDB.

Conclusion:

In this project, we successfully designed and implemented a **centralized Data Warehousing solution in Azure** to provide unified **Sales Insights** across multiple data sources. Leveraging **Azure Data Factory (ADF)**, **Azure Data Lake Storage (ADLS)**, and we established a robust end-to-end pipeline for ingestion, transformation, and reporting.

- A total of **7 Pipelines** were developed to automate the data movement and orchestration from raw sources into the target warehouse.
- **18 Datasets** were created to represent structured definitions of the source and destination data.
- **7 Dataflows** were implemented to perform necessary transformations, including data cleansing, enrichment, and business calculations such as Total Revenue, Total Cost, and Gross Profit.

The final solution ensures:

- **Scalability** – new sales files or updated dimensions are seamlessly ingested without breaking existing workflows.
- **Data Quality** – transformations guarantee consistent formats, cleaned IDs, and proper keys for modeling.
- **Business Value** – fact and dimension tables support intuitive reporting in Power BI, enabling advanced sales analysis by geography, product, category, and sales representatives.

This project demonstrates the complete **end-to-end data engineering lifecycle in Azure**, from raw data ingestion to insightful visualization, ensuring a resilient, automated, and business-ready data warehouse system.